

Algorithm 1

Binary Search (Hodgkinson)

Speech and silence segmentation

Frame 25 ms. Hop 10 ms.

Minimum silence duration: 200 ms

Name / Student ID: pending

Algorithm 1 workflow

1. Read WAV with 25/10 ms framing



2. Compute and normalize STE



3. Learn threshold by binary search



4. Create candidate speech using the learned threshold

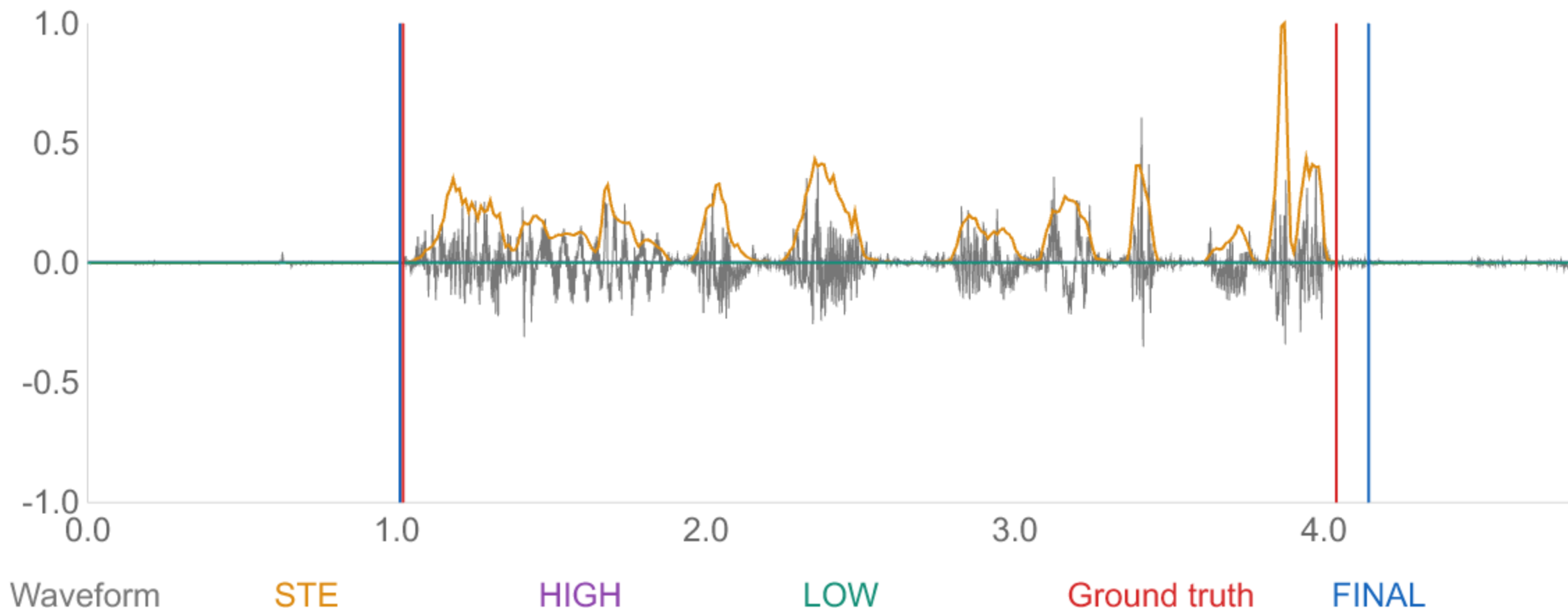


5. Hysteresis: merge <200ms gaps, discard <100ms speech

Learned threshold: 0.001029338

phone_F2: test results

MAE 57.50 ms RMSE 74.58 ms

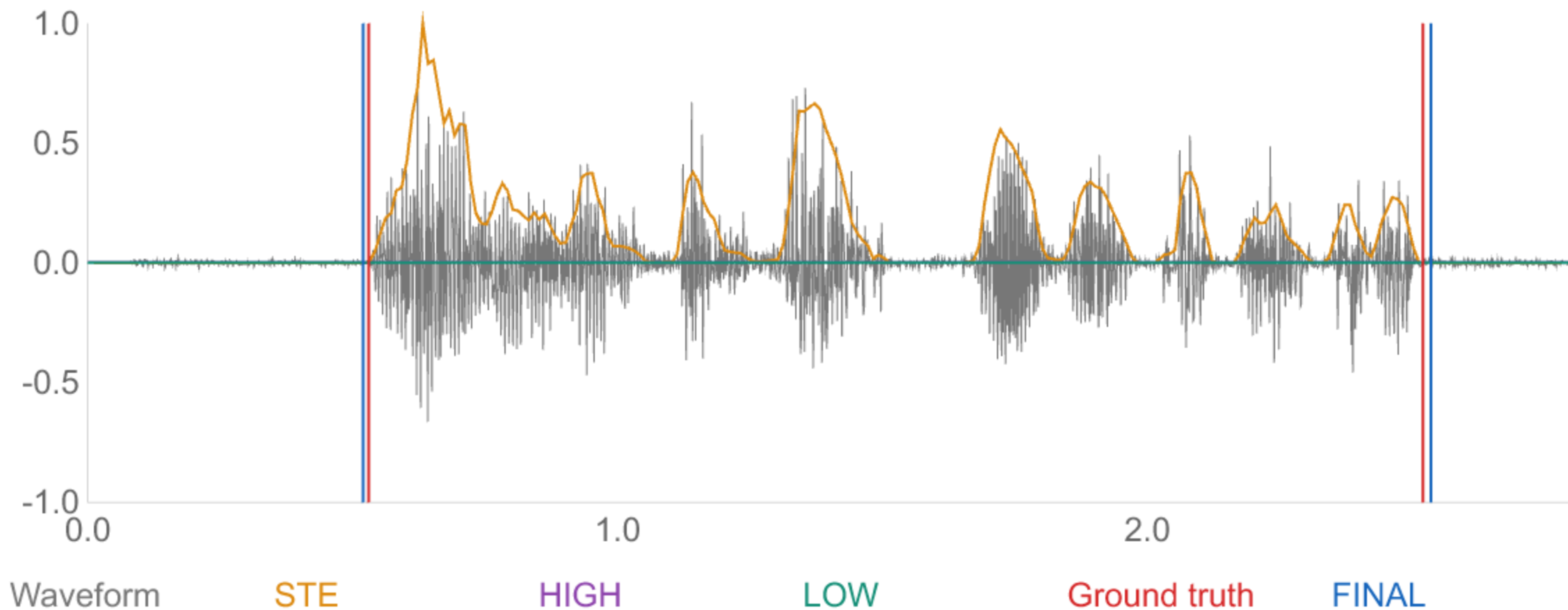


Time (s)

Noise tails can extend speech.

phone_M2: test results

MAE 12.50 ms RMSE 12.75 ms

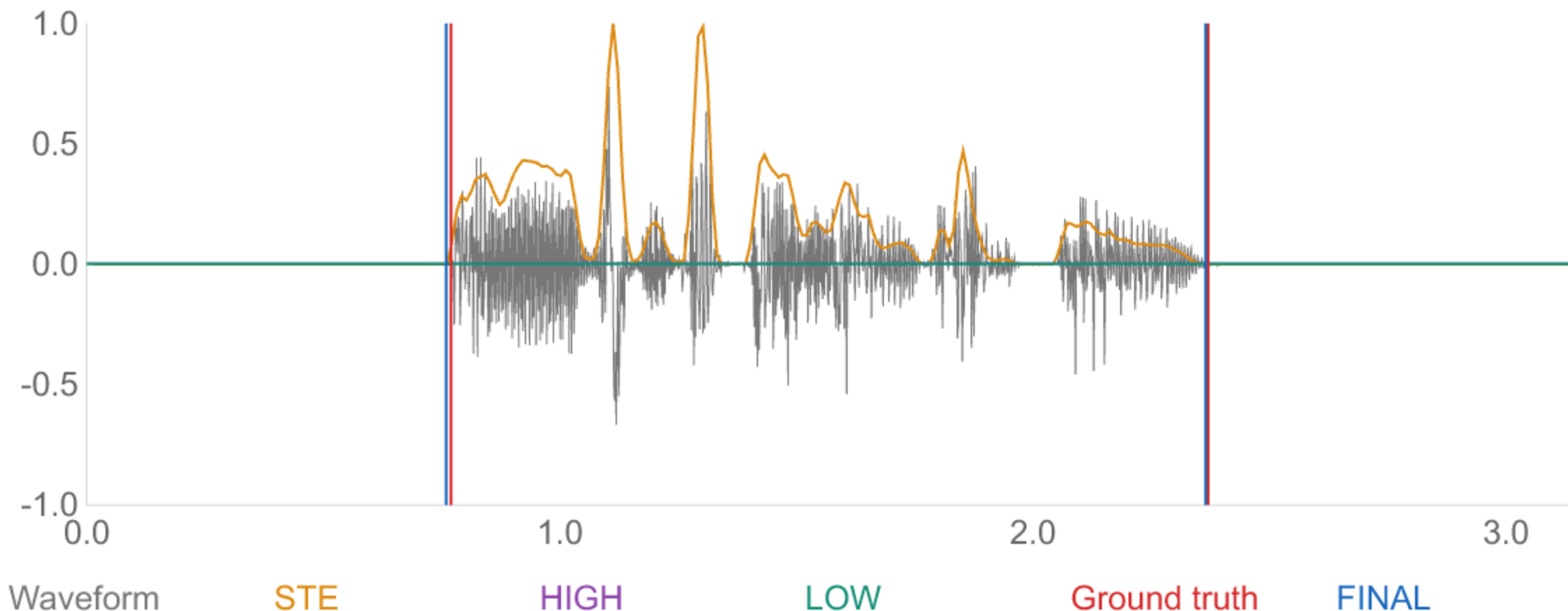


Time (s)

Noise tails can extend speech.

studio_F2: test results

MAE 7.51 ms RMSE 7.91 ms

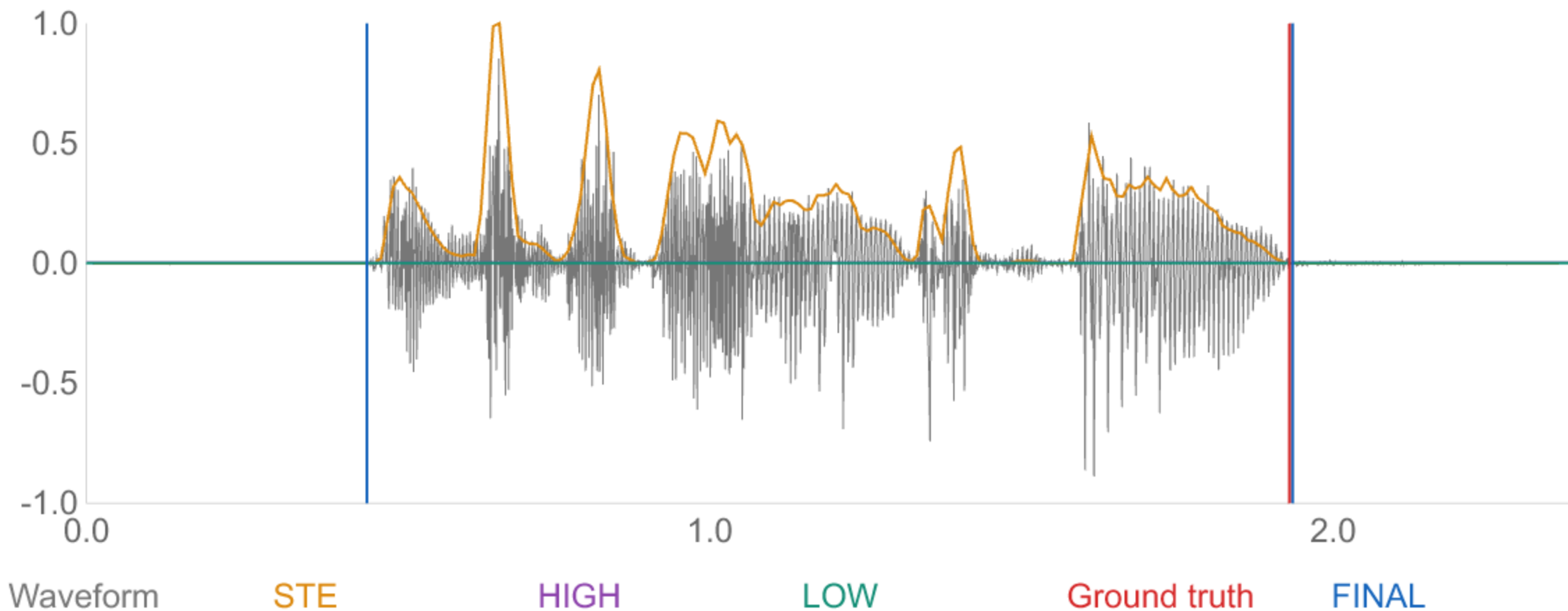


Time (s)

Framing and thresholds affect boundary error.

studio_M2: test results

MAE 2.49 ms RMSE 3.53 ms



Time (s)

Framing and thresholds affect boundary error.

Error on four test files

File	MAE (ms)	RMSE (ms)
phone_F2	57.50	74.58
phone_M2	12.50	12.75
studio_F2	7.51	7.91
studio_M2	2.49	3.53

Mean MAE: 20.00 ms

Noise extends speech tails.